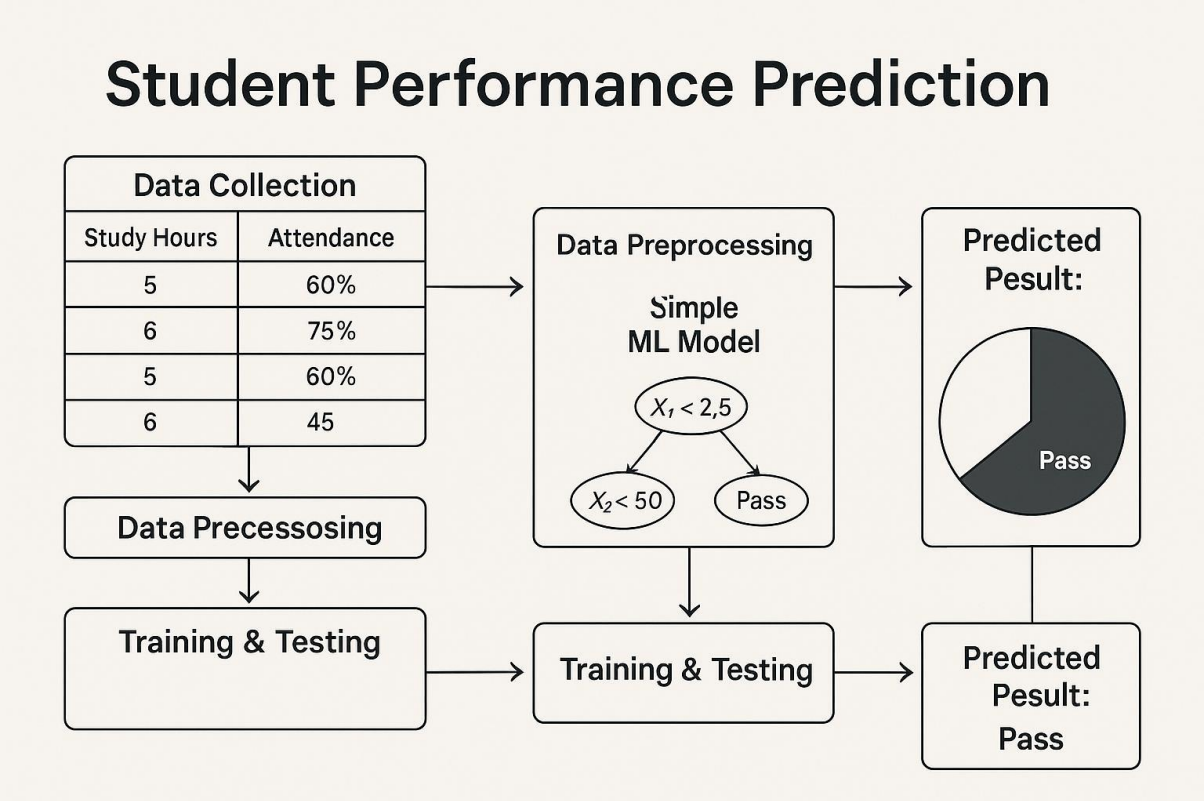
Q1. Concept Map of the Data Science Process for Student Performance

Question: A college analyzes student performance using attendance records, internal marks, assignment scores, and final exam results.

Concept map diagram reference

The generated reference image contains all five concept maps. The following sections provide the content separately so you can copy each answer into your assignment.



Concept map flow to draw

Data Sources

Attendance • Internal Marks • Assignment Scores • Final Exam Results

Data Collection

Gather records from college systems

Data Understanding

Check data types, missing values, and patterns

Data Cleaning and Integration

Correct errors and combine records using Student ID

Data Transformation

Normalize marks, convert formats, create total scores

Prepared Dataset

Ready for student performance analysis and prediction

Explanation

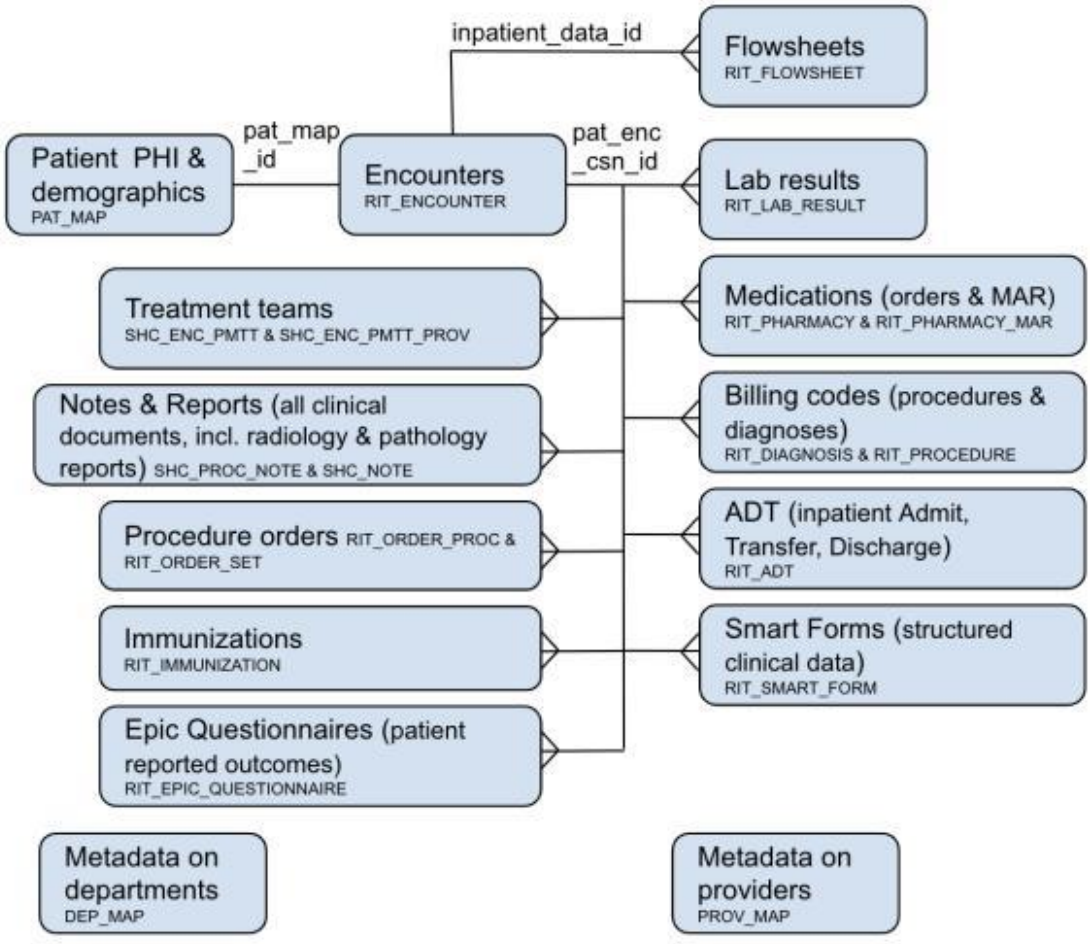
1. Data collection: The college collects attendance percentages, internal examination marks, assignment scores, and final examination results.
2. Data understanding: The collected data is examined to understand its structure, data types, and relationships.

3. Data cleaning: Missing marks, duplicate records, and incorrect values are identified and corrected.
4. Data integration: Information from different sources is combined into one dataset using a common Student ID.
5. Data transformation: Marks are converted into suitable formats, numerical values may be scaled, and new features such as total marks and attendance percentage can be calculated.
6. Final output: The prepared dataset can be used to identify students who need support, analyze academic performance, and predict results.

   Conclusion: The Data Science process converts raw college records into a clean and useful dataset for understanding and improving student performance.

Q2. Concept Map of Hospital Data Collection and Analysis

Concept map to draw



Data Collection Sources
Registration Forms
Name, age, gender, contact details
Lab Reports
Blood tests, urine tests, X-rays
Medical Devices
Heart rate, blood pressure, temperature

Doctor Notes
Symptoms, diagnosis, treatment
Data Integration
Combine patient information using Patient ID
Data Preprocessing
Clean missing values and standardize formats
Data Analysis
Disease prediction • Risk assessment • Treatment evaluation

Explanation

1. Registration forms: Collect basic patient details such as name, age, gender, address, and medical history.
2. Laboratory reports: Provide test results that help doctors understand the patient's health condition.
3. Medical devices: Record real-time information such as heart rate, blood pressure, and body temperature.
4. Doctor notes: Contain symptoms, diagnosis, prescriptions, and treatment details.
5. Data integration: Combines information from all sources into a single patient dataset, usually using a unique Patient ID.
6. Data preprocessing: Removes errors, handles missing values, and converts information into a consistent format.
7. Data analysis: Helps identify disease patterns, assess patient risks, evaluate treatments, and plan hospital resources.

Conclusion: Combining different hospital data sources provides useful information for better patient care and decision-making.

Q3. Concept Map of Data Cleaning for an Online Shopping Company

Concept map to draw



Raw Shopping Dataset

Customer details • Orders • Product details • Purchase amounts
Identify Data Problems
Missing Details
Missing names, email addresses, phone numbers
Duplicate Orders
Repeated order records
Wrong Product Entries
Incorrect names, categories, or prices
Unusual Amounts
Negative or extremely high purchase values
Apply Data Cleaning Steps
Handle Missing Values
Fill, mark as unknown, or remove where appropriate
Remove Duplicates
Verify order IDs and retain valid records
Correct Errors
Standardize product names and categories
Detect Outliers
Check unusual values and investigate
Cleaned Dataset
Accurate and reliable data for analysis

Explanation

1. Identify problems: Examine the shopping dataset to find missing customer details, duplicate orders, incorrect product information, and unusual purchase amounts.
2. Handle missing values: Fill missing numerical values using suitable methods when appropriate, use an "Unknown" category for some missing information, or remove records when justified.
3. Remove duplicates: Check order IDs and other fields to identify repeated records. Keep valid orders and remove confirmed duplicates.
4. Correct product entries: Fix spelling mistakes, standardize product names, validate prices, and correct incorrect categories using reliable reference data.
5. Detect outliers: Identify unusually high or negative purchase amounts. Investigate them before deciding whether to correct or remove them, because some unusual transactions may be genuine.
6. Validate the dataset: Check whether the cleaned data is consistent, complete enough for its purpose, and free from known errors.

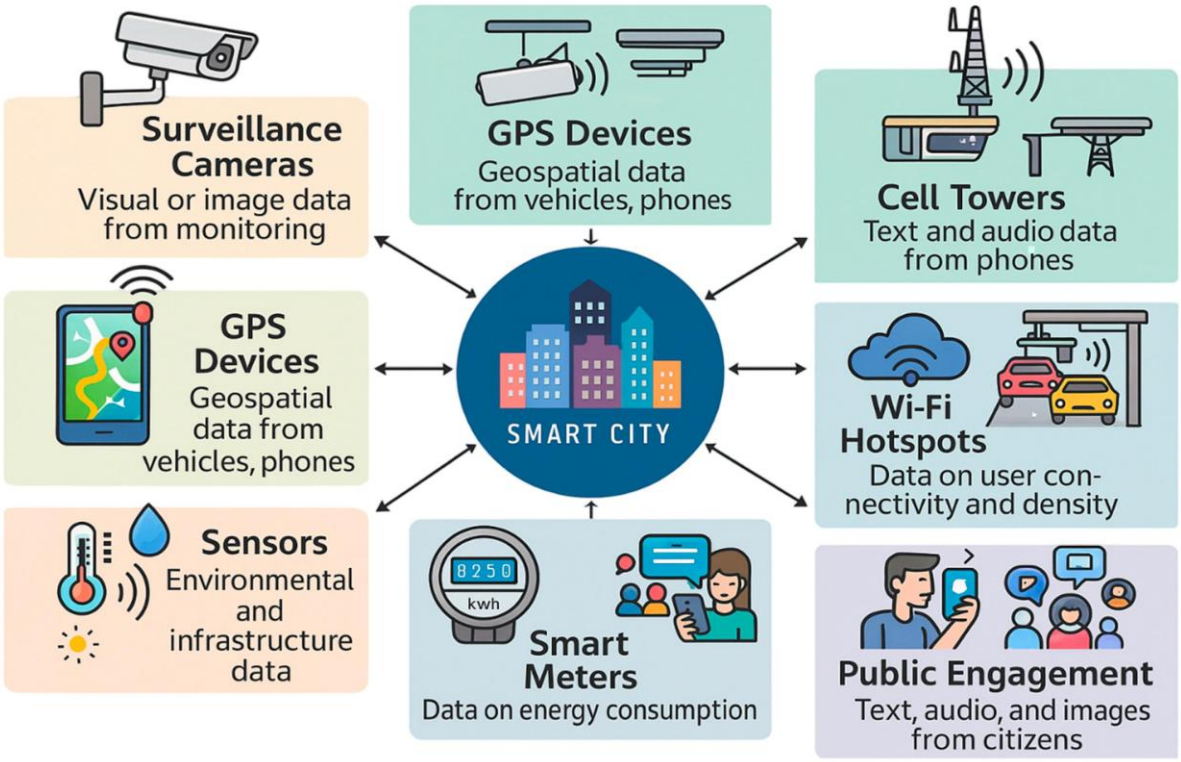
Conclusion: Data cleaning improves data quality and helps the company perform reliable sales analysis, understand customer behavior, and make better business decisions.

Q4. Concept Map of Smart City Traffic Data Integration

Concept map to draw

Multiple Data Sources
CCTV Cameras
Video, vehicle count, traffic flow
GPS Devices
Location, vehicle speed, routes

Traffic Sensors
Vehicle density, road occupancy
Mobile Applications
User reports, location, traffic updates
Data Collection
Gather real-time and historical traffic data
Data Integration
Match timestamps and locations, standardize formats, remove duplicates
Unified Traffic Dataset
One dataset containing location, time, speed, vehicle counts, and traffic conditions
Analysis and Applications
Congestion prediction • Route optimization • Smart traffic management



Explanation

1. CCTV cameras: Capture road videos and provide information about vehicle counts and traffic movement.
2. GPS devices: Provide vehicle locations, speeds, travel routes, and timestamps.
3. Traffic sensors: Measure traffic density, vehicle movement, and road occupancy.
4. Mobile applications: Provide traffic reports, user locations, and updates about road conditions.
5. Data collection: Gathers information from all sources in real time or from stored records.
6. Data integration: Converts information into common formats, synchronizes timestamps, matches locations, and removes duplicate records.
7. Unified dataset: Combines all the information into a single dataset with common fields such as time, location, speed, and vehicle count.
8. Analysis: Uses the integrated data to predict congestion, suggest better routes, and improve traffic signal management.

Conclusion: Data integration combines traffic information from different sources into one useful dataset, helping smart cities improve road safety, reduce congestion, and manage traffic efficiently.

Q5. Concept Map of Data Transformation for Bank Loan Approval Prediction

Concept map to draw

Original Customer Dataset

Income • Age • Job Type • Loan Amount • Credit Score

Data Transformation

Encoding

Convert job type into numerical values using one-hot or label encoding

Normalization

Scale numerical values to a common range, such as 0 to 1

Standardization

Convert numerical features to a common statistical scale

Other Transformations

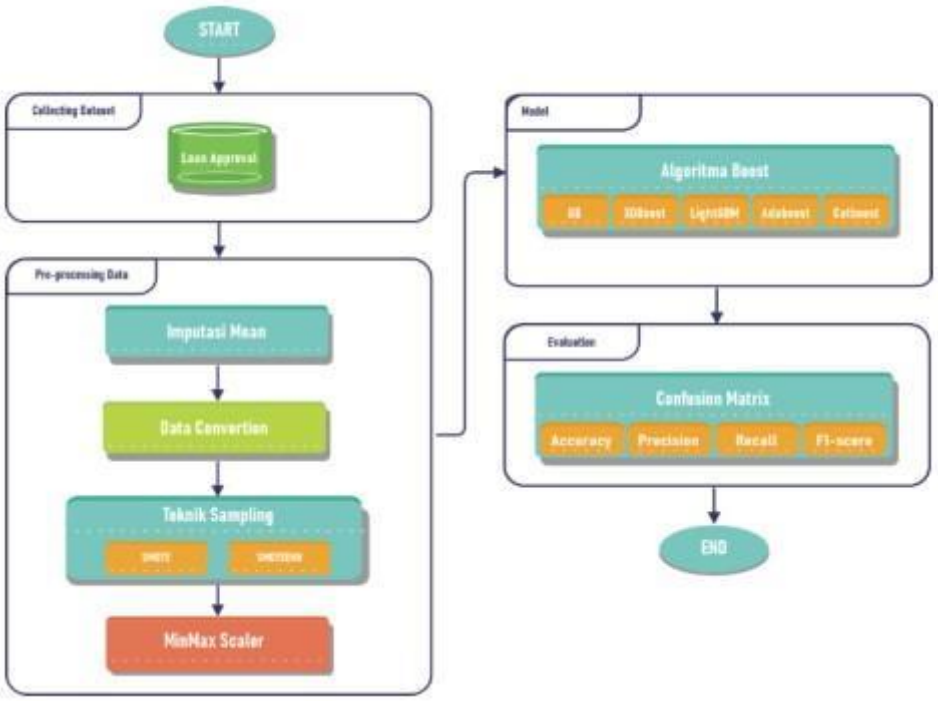
Handle missing values, outliers, and create useful features

Transformed Dataset

Numerical, consistent, and model-ready features

Loan Approval Prediction Model

Predict loan approval using historical customer and loan data



Explanation

1. Original dataset: The bank collects customer income, age, job type, requested loan amount, and credit score.
2. Encoding: Job type is categorical data, so it must be represented in a form suitable for the selected model. For example, one-hot encoding can represent Salaried, Self-employed, and Business as separate binary columns.

3. Normalization: Converts numerical features to a common range, often between 0 and 1. Min-max normalization is one common method.
   \[x'=\frac{x-x_{\min}}{x_{\max}-x_{\min}} \]
   4. Standardization: Converts values to a scale with mean 0 and standard deviation 1. It is often useful for models sensitive to feature scales.
   \[z=\frac{x-\mu}{\sigma} \]
5. Feature scaling: Ensures that features such as income, loan amount, age, and credit score do not differ excessively in scale when the selected algorithm is sensitive to these differences.
6. Other transformations: Handle missing values, investigate outliers, and create useful features, such as the ratio of loan amount to income.
7. Final dataset: The transformed features are supplied to a machine-learning model trained on historical loan decisions. The model predicts loan approval based on learned patterns, subject to the bank's validation and fairness requirements.
   Conclusion: Data transformation converts raw customer information into a consistent format that helps machine-learning models analyze loan applications more effectively.